

DAX FUNCTIONS

Aggregate Functions:

Function	Description	Use Case
SUM()	Calculates the total of a numeric column.	Calculate the total sales amount in a table. Example: SUM(Sales[Amount])
AVERAGE()	Returns the average (arithmetic mean) of a numeric column.	Find the average sales amount per transaction. Example: AVERAGE(Sales[Amount])
MIN()	Returns the smallest value in a column.	Find the lowest order amount in a table. Example: MIN(Sales[OrderAmount])
MAX()	Returns the largest value in a column.	Find the highest sales amount. Example: MAX(Sales[Amount])
COUNT()	Counts the number of non-blank values in a column.	Count the total number of transactions. Example: COUNT(Sales[OrderID])
COUNTA()	Counts the number of non-empty values in a column.	Count non-blank product names in a table. Example: COUNTA(Product[ProductName])
COUNTBLANK()	Counts the number of blank (empty) values in a column.	Identify how many records are missing data in a column. Example: COUNTBLANK(Sales[Amount])
DISTINCTCOUNT()	Counts the number of unique values in a column.	Count the number of unique customers. Example: DISTINCTCOUNT(Sales[CustomerID])
SUMX()	Calculates the sum of an expression evaluated row by row.	Sum the product of quantity and price for each row. Example: SUMX(Sales, Sales[Qty] * Sales[Price])
AVERAGEX()	Calculates the average of an expression evaluated row by row.	Calculate the average profit margin across transactions. Example: AVERAGEX(Sales, Sales[Profit]/Sales[Revenue])
MINX()	Returns the smallest value of an expression evaluated row by row.	Find the lowest calculated tax amount per transaction. Example: MINX(Sales, Sales[Amount] * 0.1)
MAXX()	Returns the largest value of an expression evaluated row by row.	Find the highest calculated profit per transaction. Example: MAXX(Sales, Sales[Profit])

Key Notes:

- SUMX(), AVERAGEX(), MINX(), MAXX()** are **iterator functions**. They allow you to evaluate a custom expression row by row and then aggregate it.
- DISTINCTCOUNT()** is particularly useful for counting unique values in large datasets, such as the number of unique customers, products, or categories.
- Functions like **COUNTBLANK()** are essential for data validation and identifying missing or incomplete records.

LOGICAL FUNCTIONS

Function	Description	Use Case
IF()	Returns one value if a condition is TRUE, and another if FALSE.	Categorize sales as "High" or "Low" based on a threshold. Example: IF(Sales[Amount] > 1000, "High", "Low")
AND()	Returns TRUE if all conditions are TRUE.	Check if a sale is above \$1000 and belongs to a specific region. Example: AND(Sales[Amount] > 1000, Sales[Region] = "West")
OR()	Returns TRUE if any condition is TRUE.	Identify if a sale is either above \$1000 or belongs to a specific region. Example: OR(Sales[Amount] > 1000, Sales[Region] = "West")
NOT()	Returns TRUE for a FALSE condition and vice versa.	Exclude rows with a certain condition. Example: NOT(Sales[Region] = "East")
TRUE()	Returns the logical value TRUE.	Use as a placeholder for TRUE conditions in complex expressions. Example: SWITCH(TRUE(), ...)
FALSE()	Returns the logical value FALSE.	Use as a placeholder for FALSE conditions.
IFERROR()	Returns a specified value if an expression results in an error, otherwise returns the expression result.	Handle division by zero. Example: IFERROR(Sales[Revenue] / Sales[Quantity], 0)
SWITCH()	Evaluates an expression against multiple values and returns corresponding results.	Categorize products based on sales amount ranges. Example: SWITCH(TRUE(), Sales[Amount] > 1000, "High", "Low")
CONTAINS()	Returns TRUE if a table contains at least one row where all specified columns match the criteria.	Check if a specific value exists in a table. Example: CONTAINS(Sales, Sales[ProductID], "P123")
ISBLANK()	Returns TRUE if a value is blank (empty).	Identify if a column contains missing data. Example: ISBLANK(Sales[Amount])
ISERROR()	Returns TRUE if an expression results in an error.	Check if a formula returns an error. Example: ISERROR(Sales[Revenue] / Sales[Quantity])
ISEMPTY()	Returns TRUE if a table is empty.	Verify if a filtered table has no rows. Example: ISEMPTY(FILTER(Sales, Sales[Amount] > 1000))
ISNUMBER()	Returns TRUE if a value is numeric.	Validate if a value in a column is a number. Example: ISNUMBER(Sales[Discount])
ISTEXT()	Returns TRUE if a value is text.	Validate if a value in a column is text. Example: ISTEXT(Sales[Comments])

Key Notes:

- IF()** is the cornerstone of logical functions and is often used with other logical functions like **AND()** or **OR()**.
- SWITCH()** provides a cleaner alternative to nested **IF()** conditions, especially for multi-condition categorization.
- CONTAINS()** is powerful for checking membership or presence in a table.

4. **IFERROR()** is essential for error handling in complex calculations, such as preventing division by zero.
5. **ISBLANK()** and **ISEMPTY()** are useful for data validation and identifying missing records.

DATE and TIME Functions

Function	Description	Use Case
DATE()	Returns a specific date from individual year, month, and day values.	Create a date field from separate year, month, and day columns. Example: DATE(2024, 12, 25)
TODAY()	Returns the current date (without time).	Display reports filtered to the current date. Example: TODAY()
NOW()	Returns the current date and time.	Use for real-time calculations that include timestamps. Example: NOW()
YEAR()	Extracts the year from a date.	Group sales data by year. Example: YEAR(Sales[OrderDate])
MONTH()	Extracts the month from a date (as a number from 1 to 12).	Filter sales data by specific months. Example: MONTH(Sales[OrderDate])
DAY()	Extracts the day from a date.	Identify transactions occurring on a specific day of the month. Example: DAY(Sales[OrderDate])
WEEKDAY()	Returns the day of the week as a number (1 to 7).	Analyze sales patterns by weekdays. Example: WEEKDAY(Sales[OrderDate])
WEEKNUM()	Returns the week number of the year for a date.	Group sales by week. Example: WEEKNUM(Sales[OrderDate])
HOUR()	Extracts the hour from a time.	Analyze hourly trends in a time series dataset. Example: HOUR(Sales[OrderTime])
MINUTE()	Extracts the minute from a time.	Identify transactions occurring at specific minutes. Example: MINUTE(Sales[OrderTime])
SECOND()	Extracts the second from a time.	Useful for precision time-based analyses. Example: SECOND(Sales[OrderTime])
EOMONTH()	Returns the last day of the month for a specified date.	Calculate month-end balances. Example: EOMONTH(Sales[OrderDate], 0)
DATEDIFF()	Returns the difference between two dates in specified units (days, months, years, etc.).	Calculate the number of days between order and delivery. Example: DATEDIFF(Sales[OrderDate], Sales[DeliveryDate], DAY)
ADDYEARS()	Adds a specified number of years to a date.	Forecast future dates. Example: ADDYEARS(Sales[OrderDate], 5)
ADDMONTHS()	Adds a specified number of months to a date.	Calculate subscription expiry dates. Example: ADDMONTHS(Sales[OrderDate], 12)
ADDDAYS()	Adds a specified number of days to a date.	Calculate due dates for invoices. Example: ADDDAYS(Sales[OrderDate], 30)
DATESBETWEEN()	Returns a table containing dates between two specified dates.	Filter data for a specific range. Example: DATESBETWEEN(Date[Date], DATE(2024,1,1), DATE(2024,12,31))
FIRSTDATE()	Returns the first date in a column or table.	Find the earliest transaction date. Example: FIRSTDATE(Sales[OrderDate])
LASTDATE()	Returns the last date in a column or table.	Identify the most recent transaction date. Example: LASTDATE(Sales[OrderDate])

CALENDAR()	Returns a table with a continuous range of dates.	Create a date table for time intelligence. Example: CALENDAR(DATE(2023,1,1), DATE(2023,12,31))
CALENDARAUTO()	Automatically generates a date table based on the dataset.	Quickly create a comprehensive date table. Example: CALENDARAUTO()

Key Notes:

1. **DATEDIFF()** is versatile for calculating time-based intervals such as delivery times, age, or project durations.
2. **EOMONTH()** is ideal for financial reporting, especially when analyzing month-end balances or deadlines.
3. **DATESBETWEEN()**, **FIRSTDATE()**, and **LASTDATE()** are essential for time-based filtering and range analyses.
4. **CALENDAR()** and **CALENDARAUTO()** help create custom date tables for advanced time intelligence.

Text Functions

Function	Description	Use Case
CONCATENATE()	Combines two text strings into one.	Combine first and last names into a full name. Example: CONCATENATE(Customer[FirstName], Customer[LastName])
CONCATENATEX()	Combines text strings from a table, separated by a delimiter.	Create a list of product names from a table. Example: CONCATENATEX(Product, Product[Name], ", ")
LEFT()	Returns the first specified number of characters from a text string.	Extract the first three letters of a product code. Example: LEFT(Product[Code], 3)
RIGHT()	Returns the last specified number of characters from a text string.	Extract the last four digits of a phone number. Example: RIGHT(Customer[Phone], 4)
MID()	Returns a substring from a text string, starting at a specified position and length.	Extract a substring from a product code. Example: MID(Product[Code], 2, 3)
LEN()	Returns the number of characters in a text string.	Calculate the length of a customer ID. Example: LEN(Customer[ID])
SEARCH()	Finds the starting position of a substring within a text string, case-insensitive.	Find the position of "@" in an email address. Example: SEARCH("@", Customer[Email])
FIND()	Finds the starting position of a substring within a text string, case-sensitive.	Locate a specific keyword in a product description. Example: FIND("Pro", Product[Description])
REPLACE()	Replaces part of a text string with another text string, based on position.	Mask part of a phone number. Example: REPLACE(Customer[Phone], 1, 3, "****")
SUBSTITUTE()	Replaces occurrences of a substring within a text string with another string.	Replace spaces with underscores in a product name. Example: SUBSTITUTE(Product[Name], " ", "_")
UPPER()	Converts a text string to uppercase.	Standardize text to uppercase. Example: UPPER(Customer[Name])
LOWER()	Converts a text string to lowercase.	Standardize text to lowercase. Example: LOWER(Customer[Email])
TRIM()	Removes all extra spaces from a text string, leaving single spaces between words.	Clean up input data with inconsistent spacing. Example: TRIM(Customer[Address])
VALUE()	Converts a text string that represents a number into a numeric value.	Convert a text field to a number for calculations. Example: VALUE("1234")
FORMAT()	Converts a value to a text string using a specified format.	Display sales amounts as currency. Example: FORMAT(Sales[Amount], "Currency")
TEXT()	Formats a value as a text string according to specified formatting.	Format a date as "DD-MM-YYYY". Example: TEXT(Order[Date], "DD-MM-YYYY")
EXACT()	Compares two text strings and returns TRUE if they are exactly the same, case-sensitive.	Validate if two product codes are identical. Example: EXACT(Product[Code1], Product[Code2])

BLANK()	Returns a blank value.	Create a placeholder blank value for text fields. Example: BLANK()
ISBLANK()	Checks if a value is blank and returns TRUE if it is.	Identify missing or empty text fields. Example: ISBLANK(Customer[Name])

Key Notes:

1. **CONCATENATE()** and **CONCATENATEX()** are crucial for combining text from multiple columns or rows.
2. **TRIM()** and **SUBSTITUTE()** are helpful for cleaning and standardizing messy data.
3. **SEARCH()** and **FIND()** differ in case sensitivity; use them as needed for text matching.
4. **FORMAT()** and **TEXT()** are essential for controlling the display of numeric or date values as strings.
5. **UPPER()** and **LOWER()** are handy for standardizing text data for comparisons.

Filter Functions

Function	Description	Use Case
ALL()	Removes all filters from the specified column(s) or table.	Calculate the total sales ignoring category filters. Example: CALCULATE(SUM(Sales[Amount]), ALL(Sales))
ALLEXCEPT()	Removes filters from all columns in a table, except for specified columns.	Keep the filter on Region while ignoring others. Example: ALLEXCEPT(Sales, Sales[Region])
ALLSELECTED()	Returns all rows in the context of a visual or current report filters.	Calculate totals based on the visible rows in a report. Example: ALLSELECTED(Sales)
CALCULATE()	Evaluates an expression in a modified filter context.	Calculate total sales for a specific region. Example: CALCULATE(SUM(Sales[Amount]), Sales[Region] = "East")
CALCULATETABLE()	Returns a table of values modified by the specified filters.	Create a filtered table of sales for a specific year. Example: CALCULATETABLE(Sales, Sales[Year] = 2023)
FILTER()	Returns a table that includes only rows that meet a specific condition.	Filter sales data for orders above \$1000. Example: FILTER(Sales, Sales[Amount] > 1000)
RELATED()	Returns a value from a related table, leveraging relationships.	Fetch customer names from a related Customer table. Example: RELATED(Customer[Name])
RELATEDTABLE()	Returns a table from a related table, leveraging relationships.	Fetch all orders related to a specific customer. Example: RELATEDTABLE(Orders)
KEEPFILTERS()	Ensures that filters in a CALCULATE expression are not removed.	Apply additional filters without overwriting existing ones. Example: CALCULATE(SUM(Sales[Amount]), KEEPFILTERS(Sales[Region] = "East"))
REMOVEFILTERS()	Removes filters from specified columns or tables.	Ignore all filters on a table or column. Example: REMOVEFILTERS(Sales)
VALUES()	Returns a one-column table containing distinct values from a column.	Fetch unique product categories for filtering. Example: VALUES(Sales[Category])
DISTINCT()	Returns a one-column table with unique values from a column.	Find distinct customer names. Example: DISTINCT(Sales[Customer])
CROSSFILTER()	Specifies the cross-filtering direction in relationships for calculations.	Change filtering direction between tables. Example: CROSSFILTER(Table1[Key], Table2[Key], BOTH)
USERELATIONSHIP()	Temporarily uses an inactive relationship for calculations.	Analyze sales using an alternate relationship. Example: CALCULATE(SUM(Sales[Amount]), USERELATIONSHIP(Sales[OrderDate], Calendar[Date]))
LOOKUPVALUE()	Returns a single value from a table for the specified conditions.	Find the price of a product based on its ID. Example: LOOKUPVALUE(Product[Price], Product[ID], "P123")
ISFILTERED()	Returns TRUE if the specified column is being filtered directly.	Check if a column is filtered in the current context. Example: ISFILTERED(Sales[Region])
ISCROSSFILTERED()	Returns TRUE if the column is being filtered indirectly.	Check if a column is filtered via a relationship. Example: ISCROSSFILTERED(Sales[Category])

EARLIER()	Returns the current row context in nested row context evaluations.	Calculate running totals or row-specific filters. Example: <code>CALCULATE(COUNT(Sales[ID]), FILTER(Sales, Sales[Amount] > EARLIER(Sales[Amount])))</code>
EARLIEST()	Returns the outermost row context in nested row evaluations.	Useful in multi-level row context evaluations. Example: Used in similar scenarios to <code>EARLIER()</code> .
SELECTEDVALUE()	Returns the value of a column in the current context, or blank if multiple values exist.	Fetch the selected product category in a slicer. Example: <code>SELECTEDVALUE(Sales[Category])</code>

Key Notes:

- 1. **CALCULATE()** is the most powerful filter function, allowing dynamic adjustments to filter context for calculations.
- 2. **ALL()**, **REMOVEFILTERS()**, and **KEEPFILTERS()** manage how filters are applied or ignored.
- 3. **FILTER()** is essential for defining custom row-level filters in calculations or table expressions.
- 4. **RELATED()** and **RELATEDTABLE()** leverage relationships between tables for retrieving data.
- 5. **USERELATIONSHIP()** and **CROSSFILTER()** manage relationships dynamically for complex scenarios.

Maths Functions		
Function	Description	Use Case
ABS()	Returns the absolute (positive) value of a number.	Convert negative sales figures to positive. Example: ABS(Sales[Profit])
CEILING()	Rounds a number up to the nearest multiple of a specified value.	Round up transaction amounts to the nearest 10. Example: CEILING(Sales[Amount], 10)
FLOOR()	Rounds a number down to the nearest multiple of a specified value.	Round down discounts to the nearest 5. Example: FLOOR(Sales[Discount], 5)
ROUND()	Rounds a number to the specified number of digits.	Round off prices to 2 decimal places. Example: ROUND(Sales[Price], 2)
ROUNDDOWN()	Rounds a number down towards zero, to the specified number of digits.	Truncate decimal parts for analysis. Example: ROUNDDOWN(Sales[Value], 1)
ROUNDUP()	Rounds a number up, away from zero, to the specified number of digits.	Round up total sales to the nearest whole number. Example: ROUNDUP(Sales[Total], 0)
MOD()	Returns the remainder after a number is divided by a divisor.	Group data into even and odd IDs. Example: MOD(Customer[ID], 2)
DIVIDE()	Performs division and handles divide-by-zero errors.	Calculate profit margin without risking divide-by-zero errors. Example: DIVIDE(Sales[Profit], Sales[Revenue])
INT()	Rounds a number down to the nearest integer.	Extract the whole number part of sales amounts. Example: INT(Sales[Amount])
TRUNC()	Truncates a number to an integer by removing the fractional part.	Simplify numbers for clearer reporting. Example: TRUNC(Sales[Amount])
POWER()	Returns the result of a number raised to a power.	Calculate exponential growth. Example: POWER(2, 3) (returns 8)
SQRT()	Returns the square root of a number.	Analyze root-based computations, e.g., standard deviation. Example: SQRT(Sales[Variance])

Key Notes:

- Rounding Functions:** Use ROUND(), ROUNDUP(), and ROUNDDOWN() for precise control over numeric rounding.
- Division Handling:** DIVIDE() is safer than / as it prevents divide-by-zero errors

Ranking and Sorting Functions		
Function	Description	Use Case
RANKX()	Returns the rank of a value in a table or column, with ties resolved using a specified order.	Rank products by total sales in ascending or descending order. Example: RANKX(ALL(Sales), Sales[Amount], , DESC)
TOPN()	Returns the top N rows of a table based on a specified expression and sorting order.	Find the top 5 products by revenue. Example: TOPN(5, Sales, Sales[Revenue], DESC)
ORDERBY()	Creates a table sorted by specified columns or expressions.	Sort a list of customers by name or sales amount. Example: ORDERBY(Customer, Customer[Sales], ASC)
NTHVALUE()	Returns the nth value in a column, based on a specified sorting order.	Find the 3rd highest sales value. Example: NTHVALUE(Sales[Amount], 3, DESC)

Detailed Notes on Key Functions:

1. RANKX():

- Highly flexible for ranking values in a column.
- Handles tie-breaking and sorting order.
- **Example:** Rank employees by performance scores.
RANKX(ALL(Employee), Employee[Score], , DESC, Skip)
- Modes:
 - **Dense:** Consecutive ranking even for ties.
 - **Skip:** Skips rank for ties.

2. TOPN():

- Filters and returns the top N rows.
- Combines well with SUMX() or other aggregate functions.
- **Example:** Get the top 3 regions by profit.
TOPN(3, Regions, Regions[Profit], DESC)

3. ORDERBY():

- Used in advanced scenarios like custom tables or as part of other functions.

- **Example:** Sort orders by quantity.

ORDERBY(Orders, Orders[Quantity], ASC)

4. NTHVALUE():

- Retrieves the nth value in a column based on the specified order.
- Useful for finding specific positions.
- **Example:** Get the second-lowest sales amount.

NTHVALUE(Sales[Amount], 2, ASC)

Key Notes:

- **Ranking (RANKX):** Critical for leaderboard-like scenarios or sorting based on KPIs.
- **Top N Filtering (TOPN):** Perfect for identifying top performers, products, or regions.
- **Sorting (ORDERBY):** Essential for creating sorted tables or intermediate calculations.
- **Nth Value (NTHVALUE):** Useful for pinpointing specific values in sorted data.

Informational Functions		
Function	Description	Use Case
ISBLANK()	Returns TRUE if the value is blank, otherwise returns FALSE.	Check if a sales column has missing values. Example: ISBLANK(Sales[Amount])
ISERROR()	Returns TRUE if the value is an error, otherwise returns FALSE.	Identify error values in calculated columns. Example: ISERROR(Sales[Profit])
ISEMPTY()	Returns TRUE if a table is empty, otherwise returns FALSE.	Validate whether a filtered table has any data. Example: ISEMPTY(FILTER(Sales, Sales[Amount] > 1000))
ISNUMBER()	Returns TRUE if the value is a number, otherwise returns FALSE.	Verify that only numeric values exist in a column. Example: ISNUMBER(Sales[ID])
ISTEXT()	Returns TRUE if the value is text, otherwise returns FALSE.	Ensure only text entries are present in a customer name column. Example: ISTEXT(Customer[Name])
ISLOGICAL()	Returns TRUE if the value is a boolean (TRUE/FALSE), otherwise returns FALSE.	Check if a calculated column is boolean. Example: ISLOGICAL(Products[IsAvailable])
ISNONTEXT()	Returns TRUE if the value is not text, otherwise returns FALSE.	Filter out non-text values from a dataset. Example: ISNONTEXT(Products[Category])
CONTAINS()	Returns TRUE if a specified value exists in a table or column.	Check if a specific product exists in a sales table. Example: CONTAINS(Sales, Sales[Product], "Item A")
HASONEVALUE()	Returns TRUE when the column has only one distinct value, otherwise FALSE.	Validate slicer selection to ensure only one value is selected. Example: HASONEVALUE(Customer[Region])
HASONEFILTER()	Returns TRUE if the column or table has been filtered to one distinct value.	Check if a column has a single filter applied. Example: HASONEFILTER(Sales[Product])
SELECTEDVALUE()	Returns the value when there is only one value selected; otherwise, returns the default value.	Extract the current filter value from a slicer. Example: SELECTEDVALUE(Customer[Region], "All Regions")
VALUES()	Returns a single-column table of unique values from a column.	Generate a unique list of regions for dropdowns or filters. Example: VALUES(Sales[Region])
DISTINCT()	Returns a single-column table with distinct values from a column or table.	Identify distinct customer names for segmentation. Example: DISTINCT(Customer[Name])
ISINSCOPE()	Returns TRUE if the specified column is in the current row context.	Verify drill-through operations in a report. Example: ISINSCOPE(Products[Category])

Key Notes:

- **Error Handling:** Use ISERROR() to avoid breaking calculations due to invalid data.
- **Dynamic Selections:** Functions like SELECTEDVALUE() and HASONEVALUE() are critical for creating interactive and responsive Power BI reports.

- **Filtering and Context:** CONTAINS() and ISINSCOPE() are essential for validating context-specific scenarios.

Detailed Examples:

1. ISBLANK():

- **Use Case:** Find missing sales data.
- **Example:**

MissingSales = IF(ISBLANK(Sales[Amount]), "No Data", Sales[Amount])

2. CONTAINS():

- **Use Case:** Check if a product exists in the data.

ProductExists = CONTAINS(Sales, Sales[Product], "Laptop")

3. SELECTEDVALUE():

- **Use Case:** Display the currently selected region from a slicer.
- **Example:**

SelectedRegion = SELECTEDVALUE(Customer[Region], "No Region Selected")

4. HASONEVALUE():

- **Use Case:** Ensure that only one region is selected.
- **Example:**

SingleRegionCheck = IF(HASONEVALUE(Customer[Region]), "One Region Selected", "Multiple Regions Selected")

5. ISERROR():

- **Use Case:** Handle error values gracefully.
- **Example:**

SafeDivision = IF(ISERROR(DIVIDE(Sales[Profit], Sales[Revenue])), 0, DIVIDE(Sales[Profit], Sales[Revenue]))

6. VALUES():

- **Use Case:** Retrieve a list of unique categories for use in another measure or visualization.
- **Example:**

UniqueCategories = VALUES(Products[Category])

Time Intelligent Functions		
Function	Description	Use Case
TOTALYTD()	Calculates the year-to-date value for a measure, based on a specified date column.	Calculate year-to-date sales. Example: TOTALYTD(Sales[Amount], Date[Date])
TOTALQTD()	Calculates the quarter-to-date value for a measure, based on a specified date column.	Calculate quarter-to-date revenue. Example: TOTALQTD(Sales[Revenue], Date[Date])
TOTALMTD()	Calculates the month-to-date value for a measure, based on a specified date column.	Calculate month-to-date profit. Example: TOTALMTD(Sales[Profit], Date[Date])
SAMEPERIODLASTYEAR()	Returns a table that contains the corresponding dates from the previous year for the given period.	Compare current year sales with the same period in the previous year. Example: SAMEPERIODLASTYEAR(Date[Date])
PARALLELPERIOD()	Returns a table with a shifted date column for a specified number of periods.	Calculate sales from the same period, but shifted by 3 months. Example: PARALLELPERIOD(Date[Date], -3, MONTH)
DATEADD()	Shifts a date column by a specified number of intervals, like days, months, or years.	Calculate sales for the last 12 months. Example: DATEADD(Date[Date], -12, MONTH)
DATERANGE()	Returns a table with dates between two given dates.	Analyze the total sales for a specific range of dates. Example: DATERANGE(Date[Date], DATE(2023,1,1), DATE(2023,12,31))
DATE()	Returns the date for a given year, month, and day.	Create a specific date value for analysis. Example: DATE(2023, 12, 25)
YEAR()	Extracts the year from a date.	Extract the year from a sales date. Example: YEAR(Sales[SaleDate])
MONTH()	Extracts the month from a date.	Extract the month from an order date. Example: MONTH(Orders[OrderDate])
DAY()	Extracts the day of the month from a date.	Extract the day from a transaction date. Example: DAY(Sales[SaleDate])
WEEKNUM()	Returns the week number of a date, based on the week start day.	Calculate the week number of a sale date. Example: WEEKNUM(Sales[SaleDate])
HOUR()	Extracts the hour from a time-based column.	Calculate the hour of a time-based event. Example: HOUR(Orders[OrderTime])
MINUTE()	Extracts the minute from a time-based column.	Calculate the minute of a time-based event. Example: MINUTE(Orders[OrderTime])

SECOND()	Extracts the second from a time-based column.	Calculate the second of a time-based event. Example: SECOND(Orders[OrderTime])
STARTOFYEAR()	Returns the first date of the year for the given date.	Identify the first date of the year for sales analysis. Example: STARTOFYEAR(Date[Date])
STARTOFQUARTER()	Returns the first date of the quarter for the given date.	Identify the first date of the current quarter for analysis. Example: STARTOFQUARTER(Date[Date])
STARTOFMONTH()	Returns the first date of the month for the given date.	Identify the start of the month for revenue calculations. Example: STARTOFMONTH(Date[Date])
ENDOFYEAR()	Returns the last date of the year for the given date.	Identify the last date of the year for sales analysis. Example: ENDOFYEAR(Date[Date])
ENDOFQUARTER()	Returns the last date of the quarter for the given date.	Identify the end of the quarter for reporting purposes. Example: ENDOFQUARTER(Date[Date])
ENDOFMONTH()	Returns the last date of the month for the given date.	Identify the last date of the month for financial reports. Example: ENDOFMONTH(Date[Date])

Detailed Examples:

1. TOTALYTD():

- **Use Case:** Calculate the year-to-date sales.
- **Example:**

SalesYTD = TOTALYTD(Sales[Amount], Date[Date])

2. SAMEPERIODLASTYEAR():

- **Use Case:** Compare the sales of the same period in the previous year.
- **Example:**

SalesLastYear = CALCULATE(SUM(Sales[Amount]),
SAMEPERIODLASTYEAR(Date[Date]))

3. PARALLELPERIOD():

- **Use Case:** Shift the sales data by 3 months to compare it with sales from 3 months ago.
- **Example:**

SalesLast3Months = CALCULATE(SUM(Sales[Amount]),
PARALLELPERIOD(Date[Date], -3, MONTH))

4. **DATEADD():**

- **Use Case:** Calculate sales for the last 12 months.
- **Example:**

SalesLast12Months = CALCULATE(SUM(Sales[Amount]), DATEADD(Date[Date], -12, MONTH))

5. **YEAR(), MONTH(), DAY():**

- **Use Case:** Extract specific components from the date for analysis.
- **Example:**

YearOfSale = YEAR(Sales[SaleDate])

MonthOfSale = MONTH(Sales[SaleDate])

DayOfSale = DAY(Sales[SaleDate])

6. **STARTOFYEAR():**

- **Use Case:** Find the first date of the current year.
- **Example:**

StartOfYear = STARTOFYEAR(Date[Date])

7. **ENDOFMONTH():**

- **Use Case:** Find the last date of the current month.
- **Example:**

EndOfMonth = ENDOFMONTH(Date[Date])

Key Notes:

- **Time Intelligence Functions:** Essential for creating time-based aggregations (YTD, MTD, QTD, etc.), comparisons to previous periods, and calculating periods like weeks, months, or years.
- Functions like **SAMEPERIODLASTYEAR()** and **TOTALYTD()** are invaluable for performing time-based trend analysis.
- **DATEADD()** and **PARALLELPERIOD()** are great for shifting dates in reporting scenarios.

